

Sai Keerthan Kasula

Cryptography & PKI Engineer | Post-Quantum Cryptography | HSM / PKCS#11 | Cryptographic Modernization

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Professional Summary

Security engineer with 3.5 years securing India's national interbank messaging network (RBI **SFMS** — **238 banks**, 2–3 crore transactions/day), with deep expertise in PKI architecture, full X.509 certificate lifecycle management, HSM/PKCS#11 integration, and enterprise cryptographic asset governance. Hands-on PQC benchmarking: ML-DSA and ML-KEM tested at production-scale financial transaction volumes against RSA/ECC baseline. Targeting PQC- migration and cryptographic modernization roles where depth in CA operations, algorithm deprecation, CBOM inventory, and NIST-aligned PQC readiness applies to enterprise and regulated infrastructure at scale.

Core Competencies

Cryptography: AES-GCM, Triple-DES, RSA, ECC, SHA-256/SHA-3, Digital Signatures, MACs, Key Exchange, Bouncy Castle, JCA/JCE, OpenSSL, ML-DSA (FIPS 204), ML-KEM (FIPS 203), SLH-DSA (SPHINCS+), Hybrid Cryptography

PKI & Certificate Management: X.509 Certificate Lifecycle (Issuance, Renewal, Revocation, Rotation, Expiry Governance), CBOM Inventory & Cryptographic Asset Discovery, CA Hierarchy Operations, Trust Chain Validation, CRL/OCSP, ASN.1, RFC/PKCS Standards, Crypto-Agility

HSM & Key Management: IBM HSM, PKCS#11 Integration, Secure Token Management, Multi-Vendor HSM Support, HSM-Backed Signing, Key Protection Strategies

Standards & Frameworks: NIST IR 8547 (PQC Migration Roadmap), CNSA 2.0 (Algorithm Transition), CISA PQC Migration Guidance, FIPS 140-3, FIPS 203/204, Algorithm Deprecation & Sunset Planning (MD5, SHA-1, 3DES)

Security Operations: Vulnerability Assessment & Remediation, Burp Suite, NIST Framework Compliance, Secure Code Review (SonarLint, Fortify), Audit Readiness, Security Incident Response

Development: Java 8/17, Spring Boot, REST APIs, SQL, Git, Maven, Postman

Professional Experience

Tata Consultancy Services (TCS) | **Client: IFTAS, A Subsidiary of Reserve Bank of India** *Jan 2023 – Present*
Security Engineer - SFMS (Structured Financial Messaging Service)

- Managed cryptographic asset inventory across 230+ bank nodes; cataloging certificate ownership, algorithm types, key parameters (size, expiry, algorithm family), & crypto metadata across all SFMS endpoints, applications, and servers; maintained **CBOM-aligned**, audit-ready crypto asset records to support **PQC** readiness assessment and RBI compliance reporting.
- Architected and delivered zero-downtime **AES-256-GCM migration** across 238-bank national interbank network; phased backward-compatible rollout reduced deployment complexity by **50% vs. simultaneous migration**.
- Led **algorithm deprecation** and ASN.1/RFC-compliant digital signature governance across 60+ independent banks — drove full ecosystem migration off MD5WithRSA and SHA1WithRSA within a 7-month timeline through technical demonstration and operational consequence framing; mentored 10–15 associates to scale coordination across support, UAT, and production; all institutions including ICICI Bank, HDFC Bank, and Axis Bank achieved compliance within the planned timeline.
- Reduced **SFMS Signer** signing latency by 50% (200 ms → 100–120 ms, production-validated) – diagnosed that **SunPKCS#11** cannot surface native token-management calls; built a custom PKCS#11 integration layer providing automatic HSM slot detection and certificate metadata caching; resolved Java 17 strong-encapsulation constraints and eliminated manual slot-ID entry for 230+ bank operators.
- Initiated and led internal PQC readiness benchmarking — tested ML-DSA (FIPS 204) and ML-KEM (FIPS 203) against RSA/ECC baseline at production-scale financial transaction volumes using Bouncy Castle; identified that **PQC migration is a PKI ecosystem problem** (CA infrastructure, HSMs, certificate storage), not an algorithm replacement; framed hybrid migration strategy aligned with NIST IR 8547 migration roadmap and CNSA 2.0 transition timelines; presented findings and roadmap to **TCS NGM India** senior leadership.
- Supported enterprise CA operations – managing **CRL/chain validation**, **HSM-backed signing** token operations, and certificate authority configuration across internal PKI infrastructure serving India's national interbank settlement network.

Projects

Secure Hybrid Crypto Engine

Crypto-agility prototype unifying classical and post-quantum algorithms – RSA, ECDSA, AES-GCM, ML-KEM (FIPS 203), and ML-DSA (FIPS 204) – demonstrating hybrid migration models for regulated financial infrastructure.

GitHub: github.com/sai-keerthan/secure-hybrid-crypto-engine

Education

Chaitanya Bharati Institute of Technology

Jun 2020 – Aug 2022

Master of Computer Applications (MCA) | CGPA: 8.55 / 10.0

Awards & Publications

- TCS Beyond Excellence Award** – mandating ASN.1/RFC-compliant digital signatures across the SFMS ecosystem, strengthening cryptographic standards across RBI's national banking network (238 banks).
- Academic Excellence Award** – outstanding academic performance throughout the MCA program.
- Technical author on cryptography, PKI architecture, and secure backend development: keerthankasula.hashnode.dev